

Signal Acquisition set-up of FORS-EMG dataset

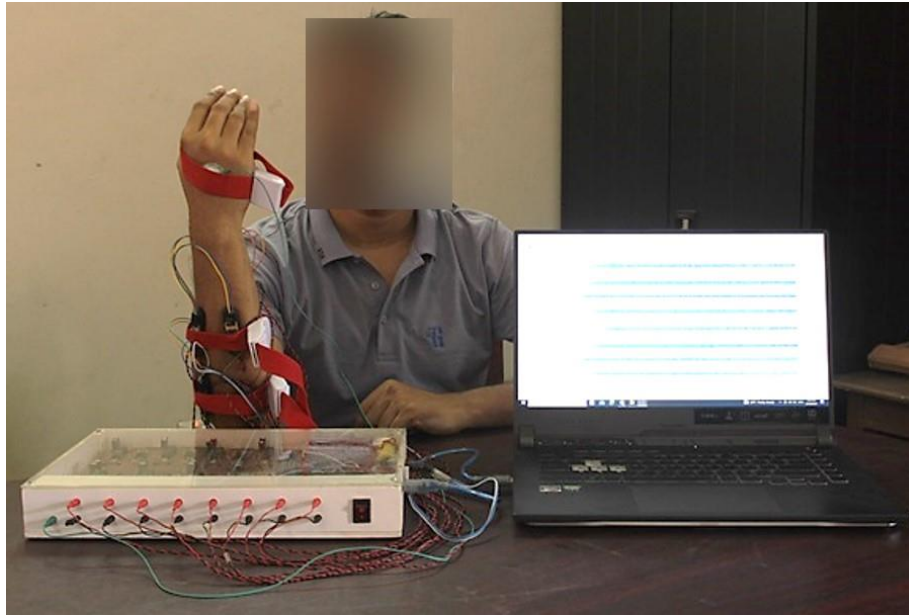


Fig. 1. Signal Acquisition setup

A specialized experimental setup was designed for sEMG signal acquisition, as depicted in Fig. 1. The device was configured with eight channels, enabling the recording of sEMG signals to monitor forearm muscle activity in a single subject.

Each channel was sampled at a frequency of 985 Hz. The system utilized eight active electrodes, with four (modified from the MFI bar electrode, MFI Medical Equipment, Inc., USA) labeled as CH1 to CH4, placed circumferentially around the elbow, and the remaining four, labeled as CH5 to CH8, positioned around the middle of the forearm, as illustrated in Fig. 2. The common ground electrode (MFI Medical Equipment, Inc., USA) was placed on the dorsal side of the hand, as shown in Fig. 2.

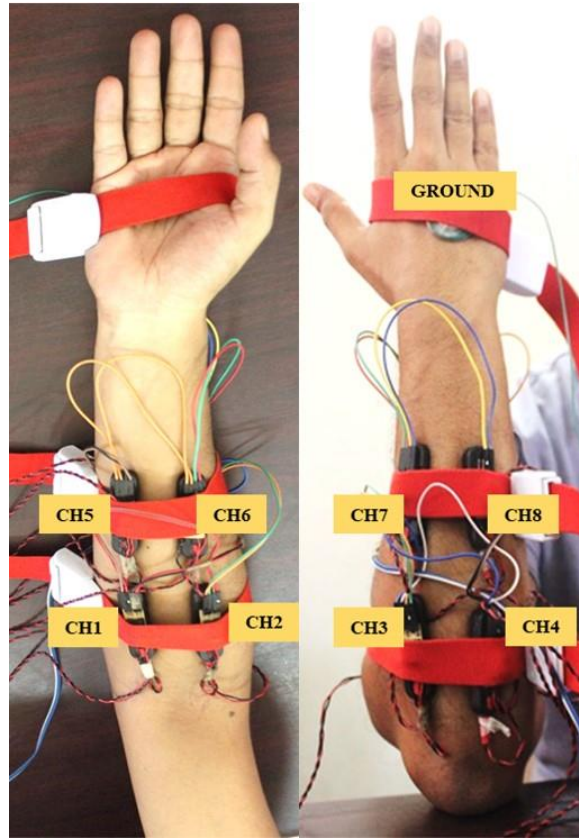


Fig. 2. Electrode Placements on elbow and forearm sites

Two electrodes (CH1, CH2) were placed on the flexor digitorum superficialis and flexor carpi radialis muscles in the anterior compartment of the forearm, just proximal to the elbow joint. On the posterior side of the forearm, electrodes CH3 and CH4 were positioned over the extensor digitorum and flexor carpi ulnaris muscles.

At the mid-forearm level, CH5 and CH6 were aligned with CH1 and CH2, respectively, targeting the same anterior forearm muscles. Similarly, CH7 and CH8 were aligned with CH3 and CH4, focusing on the corresponding muscles on the posterior side of the forearm.